Lecture 07

04.11.2025

Back to the example $u_t(t,x) - \Delta u(t,x) = f(t,x)$

Let us for example assume that $x \in (0,1)$ and $t \in (0,T)$ for some $T > 0$.

If we specify a (homogeneous) Dirichlet boundary condition

$$u(t,0) = 0 = u(t,1) \quad \text{for all } t \in (0,T)$$

and an initial condition (i.e. a Dirichlet condition on $\{0\} \times (0,1)$)

$$u(0,x) = g(x) \quad \text{for all } x \in (0,1)$$

$\Rightarrow$ our previous arguments for existence of multiple solutions no longer works since for $v(t,x) = u(t,x) + c$ and a solution u we have

$$v(0,x) = u(0,x) + c = g(x) + c \neq g(x) \text{ if } c \neq 0$$

Rule of thumb for boundary/initial conditions:

- elliptic: PDE + boundary conditions
- parabolic: PDE + boundary conditions + one initial condition
- hyperbolic (2nd order PDE): PDE + boundary conditions + two initial conditions

BUT: this rule of thumb does not automatically yield uniqueness, cf. example of homogeneous Neumann boundary conditions from exercises!

Finite difference methods: Neumann boundary conditions, 2D problems and convergence

Recall the FD scheme for the 1D Poisson equation

- Given $\Omega = (a,b)$, $f \in C(\Omega)$, $g_a, g_b \in \mathbb{R}$
- Find $u \in C^2(\Omega) \cap C(\overline{\Omega})$ such that $-u''(x) = f(x) \;\; \forall x \in \Omega$

$$u(a) = g_a, \; u(b) = g_b$$

Equidistant mesh: $x_0 = a < x_1 < \dots < x_{N-1} < x_N = b$, $h = \frac{b-a}{N}$

$\rightsquigarrow$ reduced linear system $L_h u_h = f_h$ where $L_h \in \mathbb{R}^{(N-1)\times(N-1)}$, $u_h, f_h \in \mathbb{R}^{N-1}$

$$\frac{1}{h^2}\begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-2} \\ u_{N-1} \end{pmatrix} = \begin{pmatrix} f(x_1) + \frac{1}{h^2} g_a \\ f(x_2) \\ \vdots \\ f(x_{N-2}) \\ f(x_{N-1}) + \frac{1}{h^2} g_b \end{pmatrix}$$

Already shown: L_h is a symmetric positive definite matrix!

$\Rightarrow$ we have existence of a <u>unique</u> solution to the discrete problem $L_h u_h = f_h$

What do we do for other types of boundary conditions, for example

$$-u''(x) = f(x) \quad \forall\, x \in \Omega \qquad u'(a) = g_a \,,\; u'(b) = g_b$$

$\rightarrow$ we can for example use one-sided difference quotients for the boundary nodes:

$$g_a = u'(a) = \lim_{\varepsilon \to 0} \frac{u(a+\varepsilon) - u(a)}{\varepsilon} \approx D^+ u(a) = \frac{u(a+h) - u(a)}{h} = \frac{u_1 - u_0}{h}$$

$$g_b = u'(b) = \lim_{\varepsilon \to 0} \frac{u(b) - u(b-\varepsilon)}{\varepsilon} \approx D^- u(b) = \frac{u(b) - u(b-h)}{h} = \frac{u_N - u_{N-1}}{h}$$

These two equations yield $\quad u_0 = u_1 - h \cdot g_a \quad$ and $\quad u_N = u_{N-1} + h \cdot g_b$

We then arrive at the discretized linear system

$$-\frac{1}{h^2}\begin{pmatrix} -1 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -1 \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ \vdots \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} -1/h \; g_a \\ f(x_1) \\ \vdots \\ \vdots \\ f(x_{N-1}) \\ 1/h \; g_b \end{pmatrix}$$

<u>Remark</u>
- the value of u_0, u_N is no longer exactly prescribed and we therefore did not eliminate these nodes from the system
- the matrix is no longer positive definite since $(1,1,\dots,1)\, L_h \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = 0$!

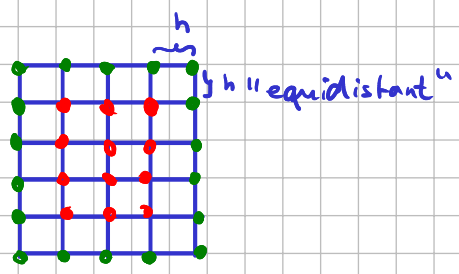
$\rightsquigarrow$ in fact this is the discrete analog of a constant function being "not seen" by the differential operator $-u''$ and u' ...

$\rightsquigarrow$ we would have to impose add. conditions to obtain a unique solution, e.g., enforcing the continuous solution to satisfy

$$\int_a^b u(x)\,dx = c \quad \text{where } c \in \mathbb{R} \text{ fixed}$$

How do we proceed in the 2D-case?

$$-\Delta u(x) = f(x) \quad \text{in } \Omega = (0,1)^2 \quad \text{(unit square for simplicity)}$$
$$u(x) = g(x) \quad \text{on } \partial\Omega$$



<u>Step 1</u> discretize Ω by a mesh Ω_h $\left(h = \frac{1}{N}\right)$

$$\Omega_h = \{(kh, \ell h) \mid k, \ell = 1, \dots, N-1\}$$
$$\overline{\Omega}_h = \{(kh, \ell h) \mid k, \ell = 0, \dots, N\}$$

$$\Gamma_h = \overline{\Omega}_h \setminus \Omega_h \Leftrightarrow \overline{\Omega}_h = \Omega_h \cup \Gamma_h$$

<u>Step 2</u> approximate $\Delta u = u_{x_1x_1} + u_{x_2x_2}$ by finite difference operators

For this, fix x_2 and consider $x = \binom{x_1}{x_2} \in \Omega_h$, define $w_1(x_1) := u(x_1, x_2)$

$$\Rightarrow D^- D^+ w_1(x_1) = \frac{w_1(x_1+h) - 2w_1(x_1) + w_1(x_1-h)}{h^2} = w_1''(x_1) + \mathcal{O}(h^2)$$

if $u \in C^4(\overline{\Omega})$, implying $w_1 \in C^4([0,1])$

Similarly for $w_2(x_2) := u(x_1, x_2)$:

$$D^- D^+ w_2(x_2) = \frac{w_2(x_2+h) - 2w_2(x_2) + w_2(x_2-h)}{h^2}$$

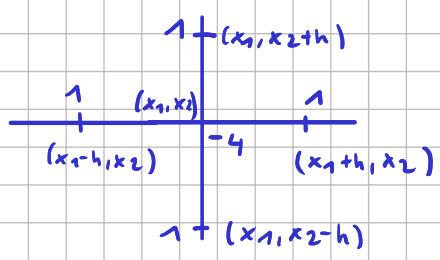
$$\Rightarrow \Delta u = u_{x_1x_1} + u_{x_2x_2} = w_1''(x_1) + w_2''(x_2) \approx D^- D^+ w_1(x_1) + D^- D^+ w_2(x_2)$$

$$= \frac{w_1(x_1+h) - 2w_1(x_2) + w_1(x_1-h)}{h^2} + \frac{w_2(x_2+h) - 2w_2(x_2) + w_2(x_2-h)}{h^2}$$

After re-substitution ($w_1, w_2 \rightsquigarrow u$) we obtain the following approximation

$$\Delta u \approx \frac{u(x_1+h, x_2) + u(x_1, x_2+h) - 4u(x_1, x_2) + u(x_1-h, x_2) + u(x_1, x_2-h)}{h^2}$$

$\Rightarrow$ so-called "5-point stencil for the 2D-Laplace"



Step 3 derive a linear system of equations: consider $u_h : \overline{\Omega_h} \to \mathbb{R}$ approximating u on Ω_h, $u_h(x) \approx u(x) \;\forall\, x \in \Omega_h$

Notation: $u_{k,l} := v_h(kh, lh)$, $k, l = 0, \dots, N$

$f_{k,l} := f(kh, lh)$, $k, l = 1, \dots, N-1$, $g_{k,l} := g(kh, lh)$, $(kh, lh) \in \Gamma$

$$\Rightarrow \; -\frac{1}{h^2}\left(u_{k+1,l} + u_{k,l+1} - 4u_{k,l} + u_{k-1,l} + u_{k,l-1}\right) = f_{k,l}, \quad k, l = 1, \dots, N-1$$

$$\underline{u_{k,l} = g_{k,l}} \qquad (kh, lh) \in \Gamma$$

$\Rightarrow$ $(N+1)^2$ unknowns $u_{k,l}$ for $k, l = 0, \dots, N$

We can eliminate nodes on the boundary by the Dirichlet b.c. to obtain

$$L_h v_h = f_h \quad \text{where } L_h \in \mathbb{R}^{(N-1)^2 \times (N-1)^2}, \; v_h \in \mathbb{R}^{(N-1)^2}, \; f_h \in \mathbb{R}^{(N-1)^2}$$

Here v_h and f_h are <u>vectors</u> representing <u>two-dimensional</u> data!

For this purpose, we use the following index mappings

- $i_{k,l} = 1 + k + l(N+1)$ for $v_h : \overline{\Omega}_h \to \mathbb{R}$
- $i_{k,l} = k + (l-1)(N-1)$ for $v_h : \Omega_h \to \mathbb{R}$

<u>Example:</u> $g = 0$, $N = 4$, $h = \frac{1}{4}$

$$-\frac{1}{h^2}\left(u_{21} + u_{12} - 4u_{11}\right) = f_{11} \qquad -\frac{1}{h^2}\left(u_{31} + u_{22} - 4u_{21} + u_{11}\right) = f_{21}$$

$$-\frac{1}{h^2}\left(u_{32} - 4u_{31} + u_{21}\right) = f_{31} \qquad -\frac{1}{h^2}\left(u_{22} + u_{13} - 4u_{12} + u_{11}\right) = f_{12}$$

$$-\frac{1}{h^2}\left(u_{32} + u_{23} - 4u_{22} + u_{21} + u_{12}\right) = f_{22} \quad \dots$$

$$\rightarrow \; -\frac{1}{h^2}\left(\begin{array}{ccc|ccc|ccc}
-4 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & -4 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 1 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\
\hline
1 & 0 & 0 & -4 & 1 & 0 & 1 & 0 & 0 \\
0 & 1 & 0 & 1 & -4 & 1 & 0 & 1 & 0 \\
0 & 0 & 1 & 0 & 1 & -4 & 0 & 0 & 1 \\
\hline
0 & 0 & 0 & 1 & 0 & 0 & -4 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 1 & -4 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & -4
\end{array}\right)
\begin{pmatrix} u_{11} \\ u_{21} \\ u_{31} \\ \hline u_{12} \\ u_{22} \\ u_{32} \\ \hline u_{13} \\ u_{23} \\ u_{33} \end{pmatrix}
=
\begin{pmatrix} f_{11} \\ f_{21} \\ f_{31} \\ \hline f_{12} \\ f_{22} \\ f_{32} \\ \hline f_{13} \\ f_{23} \\ f_{33} \end{pmatrix}$$